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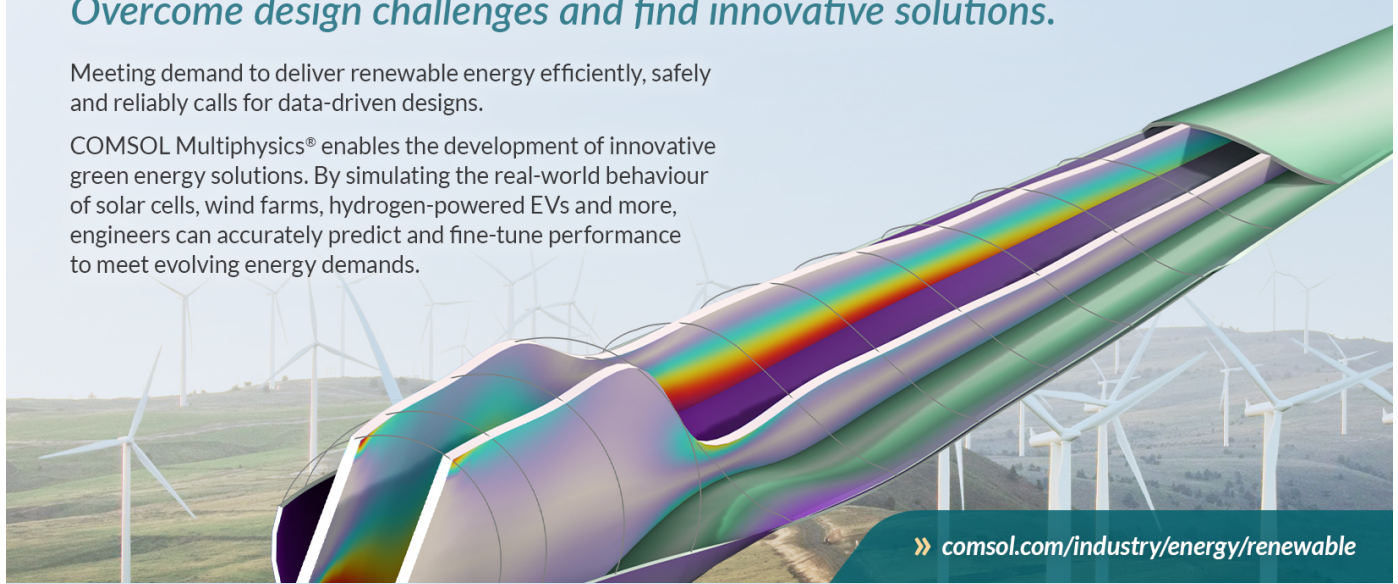
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X-ray absorption spectroscopy—a beginner's guide

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Abstract

X-ray absorption spectroscopy (XAS) has become a central tool for probing the local electronic and atomic structure of materials across physics, chemistry, and materials science, particularly under realistic operating conditions. The development of advanced synchrotron light sources, along with more affordable laboratory-based x-ray equipment, has made XAS increasingly accessible. Despite this progress, some hurdles remain for a newcomer to make use of this technique, namely the need for experimental considerations and the complexity of data curation, analysis and interpretation. This article provides a beginner-oriented introduction to XAS, with the aim of offering practical and conceptual guidance for first-time users. After introducing the fundamental principles of XAS, the distinction between x-ray absorption near-edge structure and extended x-ray absorption fine structure is outlined, and the information each regime provides on oxidation state, coordination environment, and local structure is discussed. Common experimental geometries and detection modes, including transmission, fluorescence, and electron yield, are reviewed with emphasis on their applicability and limitations. Typical data processing and analysis workflows are then summarised, with highlights on common pitfalls and best practices. Throughout, practical considerations relevant to experimental design and interpretation are emphasised, providing a foundation for readers seeking to apply XAS as an almost-routine characterisation tool.

Glossary

- **XANES** (x-ray absorption near edge structure): Near-edge region of an x-ray absorption spectrum, typically extending up to ~50 eV above the absorption edge, sensitive to oxidation state, coordination geometry, and electronic structure.
- **EXAFS** (Extended x-ray absorption fine structure): Oscillatory fine structure observed at higher energies beyond the absorption edge, arising from photoelectron scattering by neighbouring atoms and providing information on local atomic distances and disorder.
- **Core hole:** Vacant ground level electron state (s-orbital)
- **Absorption edge:** Sharp increase of the absorption coefficient at element characteristic photon energy at which a core electron is excited or ejected.
- **Pre-edge feature:** Weak spectral feature located just below the main absorption edge, often associated with electronic transitions sensitive to local symmetry and oxidation state.
- **Transmission mode:** XAS detection geometry in which the incident and transmitted x-ray intensities are measured to directly determine the absorption coefficient.
- **Fluorescence mode:** Detection of characteristic x-ray photons emitted following absorption, commonly used for dilute, highly absorbing, inhomogeneous or thin film samples.

- **Total electron yield (TEY):** Surface-sensitive detection mode based on the measurement of emitted electrons, probing the near-surface region of the sample.
- **Self-absorption:** Distortion of fluorescence XAS spectra caused by re-absorption of emitted x-ray photons within the sample, affecting spectral intensities.
- **Photoelectron wavevector (k):** Quantity related to the kinetic energy of the photoelectron in EXAFS analysis, used to describe oscillations in reciprocal space.
- **Debye–Waller factor:** Parameter describing thermal and static disorder in EXAFS analysis, reflecting the mean-square relative displacement of atomic pairs.

1. Introduction

Many key properties of materials originate from the local arrangement of atoms and the energetic distribution of their electrons. From functional oxides and catalysts to electronic and energy materials, their key chemical and physical properties often depend not only on the long-range crystal structure, but also on the local chemical environment around specific elements and their effective oxidation states.

X-ray absorption spectroscopy (XAS) stands out as a technique to assess this information. It is based on the interaction of x-rays with matter and measures the absorption of x-rays by a material as a function of the photon energy. When the incident x-ray energy is tuned across the element specific core level electron binding energy, such electron is excited into unoccupied states, either bound near the edge or continuum states at higher energy (known as the photoelectric effect), leaving a core hole behind that is subsequently filled via relaxation processes, figure 1(a). This is reflected by a sharp rise of the absorption coefficient (i.e. absorption edge in the spectrum) with characteristic features that reflect not only the electronic state of the absorbing element, but also its local atomic arrangement. Because each chemical element has absorption edges at well-defined energies, XAS provides intrinsic elemental selectivity, even in chemically complex or disordered or diluted systems.

Unlike diffraction-based methods, which primarily probe long-range periodic order, XAS is sensitive to the short-range local environment of up to approximately 5 Å. This is due to the nature of the photoelectron scattering from neighbouring atoms, which results in interference effects that create the oscillatory features which can be exploited for local structural analysis, see figure 1(b). This makes the technique particularly valuable for studying amorphous or nanostructured materials, as well as solid solutions. Indeed, XAS can be applied to samples in a wide range of physical forms, including powders, thin films, and liquids, and under diverse experimental conditions such as variable temperature, applied magnetic fields, charge currents, or controlled atmospheres to monitor chemical and/or structural changes under realistic conditions. This is particularly useful to evince transient or metastable states which are otherwise inaccessible.

An additional asset of XAS is the possibility to couple it quasi simultaneously with other spectroscopic or scattering techniques such as x-ray diffraction, small-angle x-ray scattering, x-ray emission spectroscopy, microscopy, Raman spectroscopy, etc. which offer complementary material information. [1–4]

The x-ray absorption spectrum is commonly divided into two energy regions, as depicted in figure 1(c). The near-edge region, often referred to as x-ray absorption near-edge structure (XANES), is sensitive to the oxidation state, coordination geometry, and electronic structure of the absorbing atom. At higher energies, the extended x-ray absorption fine structure (EXAFS) contains information on neighbour nature, interatomic distances, coordination numbers, and structural disorder in the immediate vicinity of the absorber.

In recent decades, advances in synchrotron radiation sources, detectors, and data analysis have greatly expanded the accessibility and applicability of XAS. As a result, the technique has become a routine component of the experimental toolbox in many disciplines. Nevertheless, pitfalls and caveats in experimental set up and data exploitation need to be taken into consideration, as it will be described in following sections.

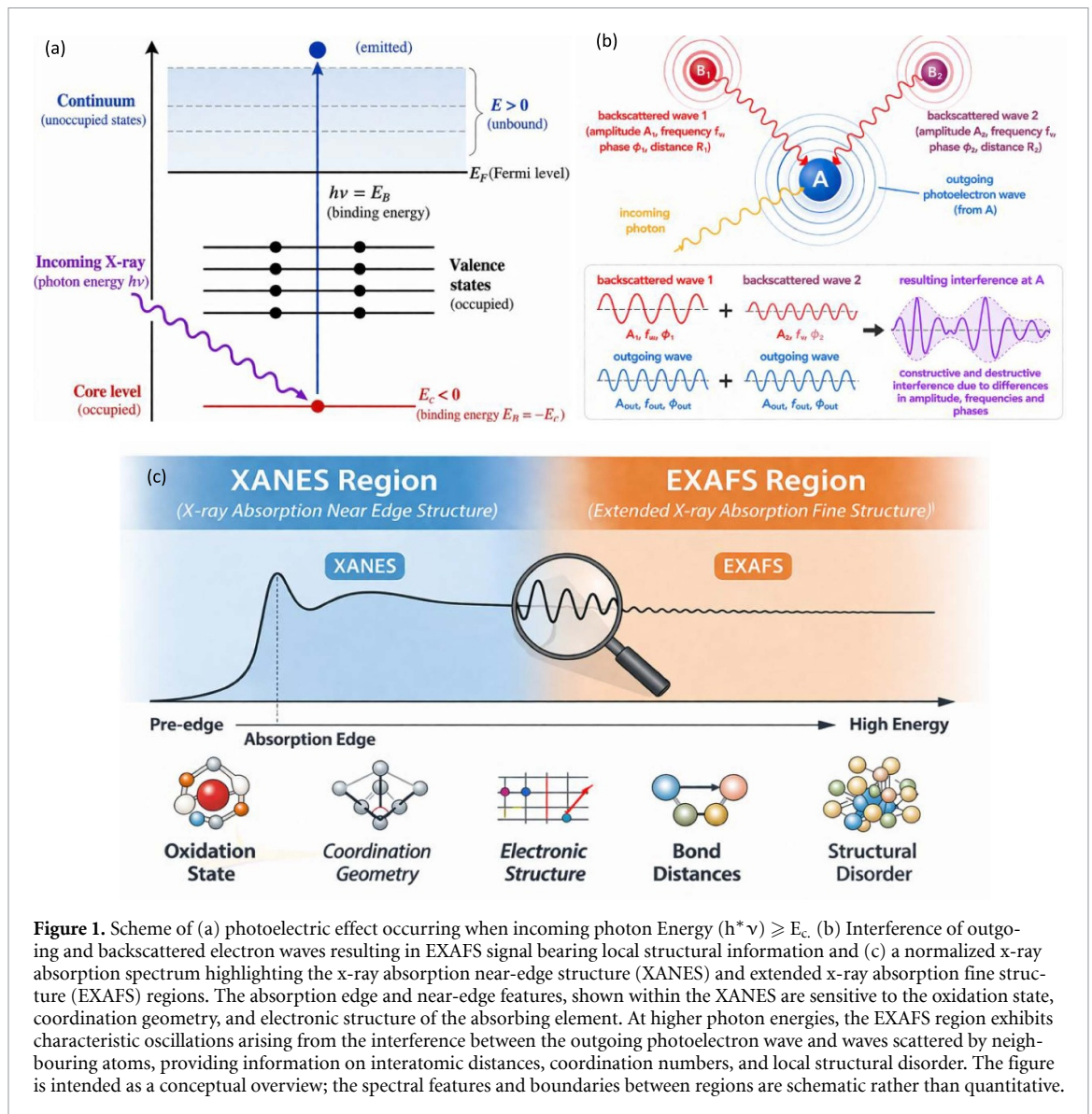


Figure 1. Scheme of (a) photoelectric effect occurring when incoming photon Energy ($h\nu \geq E_C$). (b) Interference of outgoing and backscattered electron waves resulting in EXAFS signal bearing local structural information and (c) a normalized x-ray absorption spectrum highlighting the x-ray absorption near-edge structure (XANES) and extended x-ray absorption fine structure (EXAFS) regions. The absorption edge and near-edge features, shown within the XANES are sensitive to the oxidation state, coordination geometry, and electronic structure of the absorbing element. At higher photon energies, the EXAFS region exhibits characteristic oscillations arising from the interference between the outgoing photoelectron wave and waves scattered by neighbouring atoms, providing information on interatomic distances, coordination numbers, and local structural disorder. The figure is intended as a conceptual overview; the spectral features and boundaries between regions are schematic rather than quantitative.

2. Design

The design of a XAS experiment is primarily determined by the photon energy required to access the absorption edge of interest, the detection sensitivity needed for the sample under investigation, and practical constraints related to sample form and environment. As XAS relies on scanning the incident photon energy across an elemental absorption edge, a tunable and monochromatic x-ray source is required. For this reason, most XAS experiments are carried out at synchrotron radiation facilities, which provide high photon flux and continuous and precise energy tunability over a wide energy range. Laboratory-based XAS instruments are increasingly available and can provide access to selected absorption edges, typically with lower flux and reduced flexibility, but may be suitable for routine or exploratory measurements [5].

A key design choice concerns experimental geometry and detection mode. For bulk materials with sufficient thickness and homogeneity, transmission geometry is generally preferred, as it provides a direct measure of the absorption coefficient with minimal distortion and high data quality. When transmission measurements are not feasible, for example, in the case of thin films, highly absorbing samples, inhomogeneous or dilute systems, fluorescence detection is commonly employed. In reflection-mode measurements, electron-yield and fluorescence-yield approaches can be used: total electron yield (TEY) provides high surface sensitivity (nanometre-scale depth information), whereas total fluorescence yield is more bulk-sensitive thanks to higher escape depth of photons compared to electrons (typically tens to hundreds of nanometres, strongly depending on photon energy, material composition and geometry).

Independent of the detection mode, the spatial homogeneity of the sample within the irradiated area is a critical parameter. Since XAS probes the average local environment of the absorbing element

over the illuminated volume, variations in composition, thickness, or morphology can complicate interpretation. Beam size, sample positioning and alignment should therefore be optimized to ensure representative sampling. Additional experimental parameters, such as temperature, atmosphere, or applied external fields, may be incorporated depending on the scientific question, but should be balanced against the increased experimental complexity they introduce.

3. Experiment

Once the experimental configuration has been defined, the focus shifts to the execution of the measurement and the acquisition strategy. Prior to full data collection, it is good practice to perform short test scans to confirm the approximate edge position, evaluate signal-to-noise levels, and identify potential artefacts such as monochromator glitches, detector saturation, intensity instabilities, or contributions from other elements that may impact the EXAFS signal. These preliminary scans also allow optimization of dwell times and energy step sizes appropriate for the sample and scientific objective.

Acquisition of reference spectra from well-characterized compounds containing the absorbing element is essential for energy calibration and subsequent interpretation. In transmission experiments, a reference metal foil is typically measured simultaneously with the sample by placing it between the second and third ionization chamber, ensuring consistent energy alignment throughout the experiment.

A standard XAS measurement is generally divided into three energy regions. The pre-edge region is measured to establish a baseline and assess background contributions, the near-edge region is usually acquired with finer energy steps to resolve XANES features, and the higher-energy EXAFS region is sampled more coarsely, often with longer integration times to maintain adequate counting statistics. When feasible, repeated scans under nominally identical conditions are recommended, as they enable averaging to improve data quality and provide a check for spectral reproducibility.

For *operando* and time resolved studies, the acquisition strategy must be adapted to the characteristic time scale of the process under investigation. If the system evolves faster than a full energy scan, measurements may be restricted to selected energy regions, most commonly the near-edge region, or carried out using rapid-scanning modes such as Quick-XAS, where available [6].

Throughout the experiment, real-time inspection of the acquired data quality is extremely important. Monitoring the stability of the incident intensity, verifying reproducibility between successive scans, and tracking reference spectra facilitate early detection of experimental instabilities and ensure the consistency of the dataset.

4. Data

XAS experiments generate datasets in which spectral quality, consistency, and traceability are as important as the measurement itself. During acquisition, it is therefore essential to record relevant experimental metadata together with the spectra, and to maintain a sufficiently detailed lab-book to allow reconstructing the course of the experiment at a later stage.

Before extracting structural or electronic information, raw XAS data must be pre-processed to obtain the normalised absorption coefficient $\mu(E)$. The typical workflow involves several standard steps: first, raw signals are converted into an absorption-like spectrum, for example as $\mu(E)_t = \ln(I_0/I_t)$ in transmission or approximately $\mu(E)_f = I_f/I_0$ in fluorescence mode. The resulting spectrum is then background-corrected and normalised, typically to the edge jump, to facilitate comparison between spectra. Subsequently, energy calibration is commonly carried out using reference materials measured simultaneously or intermittently to ensure consistency between scans and experimental sessions. For EXAFS analysis, conversion from energy space to photoelectron wavevector space is required, followed by background removal and weighting procedures; these steps should be applied consistently across datasets to avoid introducing systematic bias. Many different software packages are available for these such as DEMETER (now evolved into Larch), FDA, Fastosh or VIPER [7].

Data analysis strategies depend on whether information is extracted from the XANES or EXAFS regions. Near-edge spectra are often analysed using comparative approaches, such as fingerprinting against reference compounds or tracking systematic changes in edge position and spectral shape. In addition, quantitative interpretation of XANES features can be achieved through linear combination analysis of reference spectra [8] or by simulation approaches, for example using *ab initio* codes such as FDMNES [9, 10]. EXAFS analysis typically involves fitting in real or reciprocal space to

extract quantitative structural parameters, such as interatomic distances, coordination numbers and Debye-Waller factors, within meaningful physical constraints.

Operando and time-resolved measurements frequently generate large datasets with strong correlations in time or external control parameters. In such cases, statistical and multivariate analysis tools, including principal component analysis or clustering methods, can assist in identifying subtle but systematic spectral changes that may not be evident from individual scans, and reducing data dimensionality, provided their underlying assumptions and limitations are carefully considered [11].

Finally, long-term data storage and accessibility are important aspects of XAS experiments. Raw and processed data, together with calibration files and associated metadata, should be stored in well-documented and standardized formats to facilitate reproducibility, enable re-analysis as analysis methods evolve, and support data sharing within collaborations or through public repositories.

5. Practical considerations

Although XAS is conceptually straightforward, several practical aspects can significantly influence data quality and interpretation, especially for first-time users. A frequent source of artefacts originates from sample preparation. In transmission geometry, inappropriate sample thickness or non-uniform loading may result in insufficient absorption contrast or excessive attenuation, both of which complicate normalisation and quantitative analysis [12]. In fluorescence detection, low absorber concentration or spatial inhomogeneity can lead to poor signal-to-noise ratios and spectra that are not representative of the intrinsic material properties.

The choice of detection mode and experimental geometry introduces additional limitations that must be considered carefully. While fluorescence detection extends the applicability of XAS to dilute or highly absorbing samples, it is susceptible to self-absorption and saturation effects, which can distort relative spectral intensities, particularly in the XANES region. Surface-sensitive detection modes such as TEY probe only the near-surface region and may therefore yield spectra that differ substantially from bulk-sensitive measurements. Awareness of the effective probing depth is essential when comparing data acquired using different detection schemes.

In *operando* and more generally in time-resolved experiments, a common pitfall arises from a mismatch between the acquisition time of a full XAS scan and the characteristic time scale of the process under investigation. If the system evolves significantly during data collection, the resulting spectra may represent an average over multiple states rather than a physically meaningful snapshot. Restricting measurements to selected energy regions or employing rapid-scanning approaches can mitigate this issue, albeit often at the expense of reduced spectral information.

Beam-induced effects constitute another critical concern, particularly for radiation-sensitive materials. Prolonged x-ray exposure may cause local heating, induce unintended chemical reactions, or alter oxidation states, leading to apparent spectral changes that are not intrinsic to the system. Such effects are often revealed by systematic differences between successive scans or between measurements performed at different sample positions [13, 14].

Finally, caution is required during data interpretation. Over-reliance on visual fingerprinting without appropriate reference compounds, or over-parameterization in EXAFS fitting, can lead to non-physical conclusions. The information content of XAS data is inherently limited, and quantitative analysis should be performed within well-defined physical constraints, with explicit consideration of uncertainties [15].

6. Summary

XAS is a versatile and widely applicable technique for probing the local electronic and atomic structure of diverse types of materials with elemental specificity. This article provides a beginner-oriented overview of the key concepts, experimental design choices, practical aspects of measurement, and fundamental approaches to data handling and interpretation. The individual steps leading to a successful XAS are summarized in figure 2. By highlighting both the strengths of XAS and the practical considerations that govern data quality and reliability, this tutorial aims to facilitate the first steps of new users. With appropriate experimental planning, careful execution, and critical analysis, XAS can be applied as a robust and powerful characterization tool across a broad range of scientific disciplines.

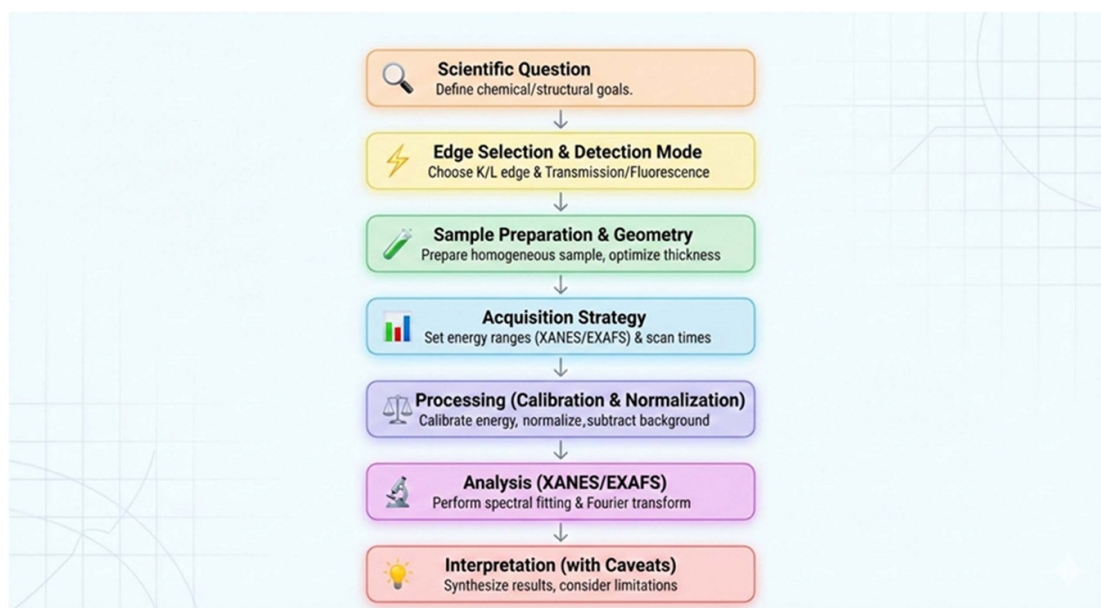


Figure 2. Workflow scheme of a typical XAS experiment. The diagram illustrates the sequential stages, from the scientific case and critical points in the experimental design and data acquisition, processing and analysis, to the final data interpretation.

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Data availability statement

All data that support the findings of this study are included within the article (and any supplementary files).

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